



Adama science and technology University

Computer science and engineering department(CSE)

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Mobile Computing and application(CSEg3306) Project.

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Complaint & Feedback Management System

Project Overview

Project Title

Complaint & Feedback Management System Mobile Application

Project Description

The Complaint & Feedback Management System is a mobile application designed to help users submit complaints and feedback digitally while enabling administrators to manage, track, and resolve complaints efficiently.

The system solves problems such as:

- Delayed responses
- Poor communication
- Lack of transparency
- Lost complaints
- Difficulty tracking complaint progress

The application allows users to:

- Submit complaints with attachments
- Track complaint status
- Receive updates and notifications
- View response history

Administrators can:

- Manage complaints
- Categorize complaints
- Update statuses
- Respond to users
- Analyze complaint trends

1. Deep Study & Analysis

1.1 Problem Statement

Many institutions and organizations still manage complaints manually using:

- verbal reports
- paper forms

- Telegram groups
- phone calls
- spreadsheets

This creates several problems:

- complaints get lost
- users receive delayed responses
- no transparency in tracking
- no centralized system
- poor accountability
- difficult reporting and analytics

The proposed system digitizes complaint handling and improves communication between users and administrators.

1.2 User Interviews

Interview Participants

Users (10)

1. University student
2. Dormitory student
3. Cafeteria customer
4. Staff member
5. Library user
6. ICT service user
7. Campus transportation user
8. Maintenance service user
9. Registrar office visitor
10. Department office visitor

Administrators (2)

1. Student service officer
2. Campus maintenance administrator

1.3 Sample Interview Questions

Questions for Users

1. How do you currently submit complaints?

2. How long does it take to receive responses?
3. Have your complaints ever been ignored?
4. Can you track complaint progress?
5. What frustrates you most about the current system?
6. Would you prefer a mobile application?
7. Would notifications help you?
8. Do you want to attach images/videos as proof?
9. What features would improve complaint handling?
10. How often do you submit complaints?

Questions for Administrators

1. How are complaints currently collected?
2. What are the major challenges?
3. What causes delays in resolution?
4. How do you organize complaints?
5. How do you communicate updates to users?
6. What type of reports are needed?
7. Would categorization help administrators?
8. What features would improve efficiency?

1.4 Interview Findings

Common Problems Identified

Users

- Complaints are ignored
- No follow-up system
- Slow responses
- No status tracking
- Difficult communication
- Repeated complaints
- Lack of transparency

Administrators

- Complaints arrive from multiple channels
- Difficult to organize records
- Manual tracking is inefficient
- Hard to prioritize urgent complaints

- Lack of reporting tools

1.5 Real-World Complaint Scenarios

Scenario 1: Dormitory Water Problem

A student reports a water shortage in the dormitory.

Current Process:

- Student tells dormitory representative
- Representative informs maintenance office
- No tracking mechanism exists
- Student does not know complaint status

Problems:

- delays
- poor communication
- repeated complaints

System Solution:

- digital complaint submission
- status tracking
- admin notifications
- resolution updates

Scenario 2: Classroom Projector Failure

A lecturer reports a broken projector.

Current Problems:

- maintenance requests are forgotten
- no priority management
- no evidence attachment

System Solution:

- categorized complaint
- image attachment
- priority management
- admin dashboard

Scenario 3: Cafeteria Hygiene Complaint

Students complain about cafeteria cleanliness.

Current Problems:

- complaints spread through social media
- difficult accountability
- delayed action

System Solution:

- centralized reporting
- complaint history
- admin monitoring
- user feedback

Scenario 4: Internet Connectivity Issue

Students report poor Wi-Fi connectivity.

Current Problems:

- multiple duplicate complaints
- no update mechanism
- difficult technician coordination

System Solution:

- categorized issue tracking
- complaint grouping
- technician assignment

Scenario 5: Registrar Office Delay

Students complain about delayed document processing.

Current Problems:

- unclear responsibility
- no response tracking
- repeated office visits

System Solution:

- transparent workflow
- response tracking

- notification system

1.6 As-Is Workflow

Current Workflow

User → Representative/Office → Administrator → Manual Processing → Resolution

Problems:

- no tracking
- no notifications
- no central database
- delays
- poor transparency

1.7 Proposed Workflow

User → Mobile Application → Database → Admin Dashboard → Status Update → Notification → Resolution

Advantages:

- centralized system
- real-time tracking
- accountability
- faster communication
- digital records

2. Requirement Specification

2.1 Functional Requirements

User Features

Authentication

- User registration
- User login/logout
- Password reset

Complaint Management

- Submit complaints
- Upload images/documents
- Edit complaints before review
- View complaint history
- Search complaints

Tracking

- Track complaint status
- Receive notifications
- View admin responses

Feedback

- Rate resolutions
- Submit additional feedback

Admin Features

Complaint Handling

- View all complaints
- Categorize complaints
- Assign priority
- Update complaint status
- Respond to complaints
- Mark complaints resolved

User Management

- Manage users
- Block malicious users
- View user activity

Analytics

- Complaint statistics
- Monthly reports
- Category analysis
- Resolution performance

2.2 Non-Functional Requirements

Performance

- Fast response time
- Efficient database queries
- Scalable architecture

Security

- Secure authentication
- Data encryption
- Role-based access

Usability

- User-friendly interface
- Responsive design
- Easy navigation

Reliability

- System availability
- Backup and recovery
- Error handling

Compatibility

- Android support
- Multiple screen sizes

Maintainability

- Modular code
- Clean architecture
- Documentation

2.3 User Stories

User Stories

Student User

- As a student, I want to submit complaints so that administrators can resolve my problems.
- As a student, I want to track complaint status so that I know the progress.
- As a student, I want notifications so that I receive updates immediately.

Administrator

- As an administrator, I want to categorize complaints so that they are organized.
- As an administrator, I want to update statuses so that users remain informed.
- As an administrator, I want analytics so that I can monitor service quality.

2.4 Use Cases

Main Use Cases

User

- Register
- Login
- Submit Complaint
- Upload Attachment
- Track Complaint
- Receive Notification
- Give Feedback

Admin

- Login
- View Complaints
- Update Status
- Respond to Complaint
- Generate Reports

3. Design Documents

3.1 System Architecture

Architecture Components

Frontend

- Flutter Mobile Application

Backend

- Fast API

Admin Panel

- Flutter Web or React Admin Dashboard

3.2 Database Schema

Entity Relationship Design

Users Table

Field	Type
User_id	string
name	string
email	string
phone	string
Role	string
Created_at	TimeStamp

Complaints Table

Field	Type
complaint_id	String
user_id	String
category_id	String
title	String
description	String
attachment_url	String
status	String
priority	String

created_at	Timestamp
updated_at	Timestamp

Categories Table

Field	Type
category_id	String
category_name	String

Responses Table

Field	Type
response_id	String
complaint_id	String
admin_id	String
response_message	String
created_at	Timestamp

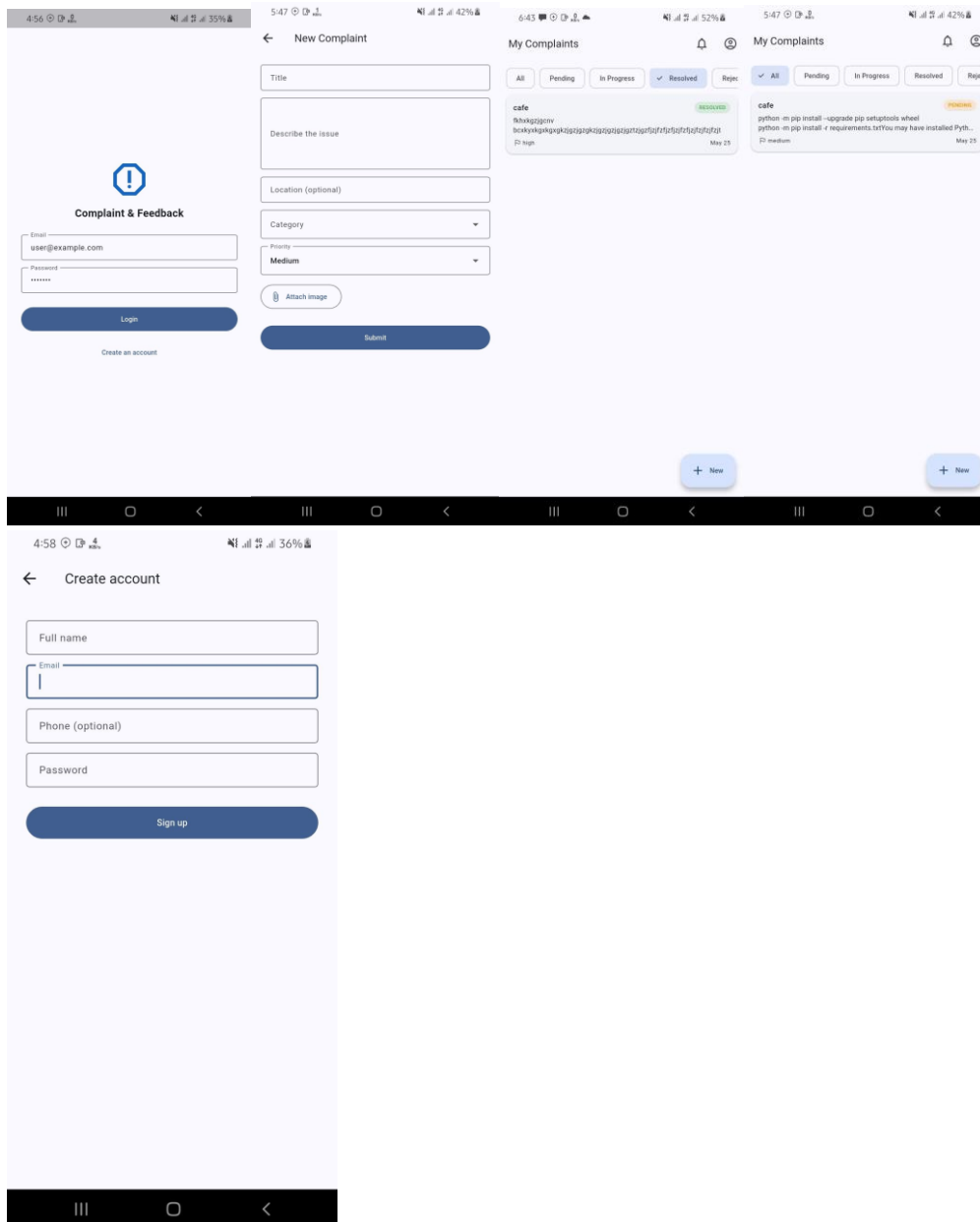
Notifications Table

Field	Type
notification_id	String
user_id	String
message	String
is_read	Boolean
created_at	Timestamp

3.3 Suggested UI Screens

User Screens

1. Splash Screen
2. Login Screen
3. Register Screen
4. Home Dashboard
5. Submit Complaint Screen
6. Complaint Detail Screen
7. Complaint History Screen
8. Notifications Screen
9. Profile Screen



Admin Screens

1. Admin Dashboard
2. Complaint List
3. Complaint Detail
4. Category Management
5. Analytics Dashboard
6. User Management

Complaints

Dashboard

Complaints

Categories

Users

Search title / description

All

✓

cafe

Demo User • Other • May 25 6:24 AM

HIGH

✗

title

Demo User • Academic • May 25 6:21 AM

MEDIUM

Dashboard

Dashboard

Complaints

Categories

Users

Total

0

Pending

0

In Progress

0

Resolved

0

Users

5

Complaints by Status

No data

Complaints by Category

Year 2018 E.C (2026 G.C)

Users

Dashboard

Complaints

Categories

Users

S

System Admin

admin@example.com

admin

D

Demo User

user@example.com

user

A

Alemu Bekele

alemu@example.com

user

S

Sara Tesfaye

sara@example.com

user

D

Daniel Mekonnen

daniel@example.com

staff

Categories

Dashboard

Complaints

Categories

Users

Name

Description

+ Add

Infrastructure

Roads, buildings, utilities

Sanitation

Garbage, cleanliness, hygiene

Academic

Class, exam, faculty related

IT Services

Network, accounts, systems

Security

Safety and security concerns

Cafeteria

Food,Water and any related issue

Other

Any Other Issue

3.4 UI/UX Design Suggestions

Color Palette

Primary Color:

- Blue or Purple

Secondary Color:

- White and Gray

Accent Color:

- Green for resolved
- Yellow for pending
- Red for urgent

Typography

Recommended Fonts:

- Poppins
- Inter
- Roboto

UX Suggestions

- Simple navigation
- Clear status indicators
- Minimal form fields
- Real-time updates
- Accessible design

3.5 API Design

Authentication APIs

POST /register POST /login POST /logout

Complaint APIs

POST /complaints GET /complaints GET /complaints/{id} PUT /complaints/{id} DELETE /complaints/{id}

Notification APIs

GET /notifications PUT /notifications/read

4. Development

4.1 Recommended Technologies

Frontend

Flutter

Backend

Firebase

Database

Cloud Firestore

Storage

Firebase Storage

Notifications

Firebase Cloud Messaging

Version Control

Git + GitHub

4.2 Flutter Project Structure

lib/ ├── main.dart ├── models/ ├── screens/ ├── services/ ├── providers/ ├──
widgets/ ├── utils/ ├── routes/ └── constants/

4.3 Core Flutter Packages

- firebase_core
- firebase_auth
- cloud_firestore
- firebase_storage
- firebase_messaging
- provider
- image_picker
- flutter_local_notifications